

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a905c5a375f92a4aa.jsonl`  
- SHA-256: `a2866de84a4976cf73ff4828413673c0ee7c6cfe5584380b90e97a2421e41872`  
- Source modified: `2026-07-02T03:01:04+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:59:38.857Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.

A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer:

```
{  
  "title": "Decision-snapshot regression track: decisions locked 06-25, zero code shipped since;  
tracker P0 row also stale",  
  "category": "tech-debt",  
  "severity": "P2",  
  "file": "docs/DECISION_SNAPSHOT_REGRESSION.md",  
  "line": 66,  
  "evidence": "$7 tracker (lines 63-73): P1-P8 all ?. Verified in code: grep 'build_stitch_plan'  
across backend/ and tests/ returns nothing (P2's pure stitch-plan seam was never factored out of  
backend/pipeline/stitcher/compiler.py), and tests/test_golden_fixtures.py has no stitch-plan  
snapshot. Minor rot: the P0 row (line 65) still shows ? '(this PR)' though the doc merged as  
#383 (commit 1283e18, 2026-06-25) and the header still says 'DRAFT'.",  
  "why_it_matters": "This gate exists precisely to bless genuine (non-AST-identical) refactors  
as quality-neutral ? i.e., exactly the class of change as the stalled #390 main.py split.  
Executing P1-P4 first would de-risk redoing the god-module split; leaving it a week-stale  
locked-decisions doc means refactors continue with no behavioral proof.",  
  "proposed_fix": "Start $7 P1+P2 (canonical decision-snapshot schema + factor pure  
build_stitch_plan() out of compiler.py) as the next Claude-doable engine work; flip the P0 row  
to ? #383 and status to IN PROGRESS in the same PR. Ideally land P1-P4 before re-cutting the  
main.py split.",  
  "effort": "L",  
  "tracked_in": "docs/DECISION_SNAPSHOT_REGRESSION.md $7 (P1-P8)"  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer,
C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-
data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator,
C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field;
non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup',
C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screencast).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited
code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent
commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete
failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it
owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training,
product/strategy/licensing, real cloud spend, counsel ToS/DPA?

HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only
commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you
personally saw the defect/debt in current code; if you cannot reproduce the claim from the code,

say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T02:59:47.516Z)

```
1      <!--
2      PROJECT DOC ? tracked design plan for a deterministic "quality-neutral proof".
3      Follows docs/PROJECT_DOC_TEMPLATE.md. $7 progress tracker is the source of truth.
4      -->
5
6      # Decision-Snapshot Regression ? a deterministic quality-neutral proof
7
8      > **Status:** `DRAFT` ? **$2 decisions LOCKED (2026-06-25); ready to start $7 P1** .
**Owner:** Avi . **Created:** 2026-06-25 . **Last updated:** 2026-06-25
9      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
10     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` once work starts.
11     > **Relation to existing docs:** **extends** `GOLDEN_REGRESSION_FIXTURES.md` (post-
detector fixtures) and the existing characterization snapshot in
`tests/test_golden_fixtures.py`; **peer of** the M4 quality-floor track and the GPU
`scripts/nightly_reelcount.py` e2e nightly.
12     > **Honesty note:** claims tagged `[verified]` (checked vs code) / `[design]`
(proposed).
13
14     ---
15
16     ## 0. TL;DR
17     A deterministic regression that proves a change is **quality-neutral** (behavior-
preserving) by freezing and comparing the pipeline's **decision output** ? the rally segment
list **+ the stitch/concat plan** ? run off a **cached trajectory** (GPU-free). It **extends**
the existing characterization snapshot (`test_replay_matches_committed_expected`, today
segmentation-only) down to the stitch plan, runs offline in CI, and gives the **behavioral
proof** that lets a *genuine* refactor (not just `ruff format`) be confidently called quality-
neutral. **Most important caveat: never hash the encoded reel `.mp4`** ? ffmpeg/x264/NVENC/mux-
timestamps make those bytes non-deterministic, so a reel-MD5 would false-alarm on every ffmpeg
bump.
18
19     ## 1. Problem & goal
20     "Presumably quality-neutral" changes (refactors) need a cheap, deterministic proof that
they actually preserve behavior ? so they can be merged (and auto-signed-off) with confidence.
Today we have two weaker tools: **AST-identity** [verified] (used to bless `ruff format` in
#379) only proves *formatting* is unchanged ? it cannot bless a real refactor (function
extraction, rename, loop reflow) that changes the AST but should preserve behavior; and the full
**real-video F1 / reel-count e2e** needs GPU + the gitignored corpus
(`scripts/nightly_reelcount.py`) [verified] so it can't gate a PR. The decision-snapshot fills
the gap: a deterministic, offline, behavior-level proof.
21
22     **What "good" looks like:** a refactor PR runs the snapshot in CI; byte-identical
decision output ? provably quality-neutral ? eligible for auto-sign-off; a behavior-changing
edit fails with a **readable diff**.
23
24     **Read first:** `tests/test_golden_fixtures.py::test_replay_matches_committed_expected`
[verified] (the existing snapshot ? its own message: *"behaviour changed vs frozen snapshot
(characterization, not correctness)"`, `backend/eval/golden_fixtures.py`,
`backend/pipeline/stitcher/compiler.py` (the concat plan), audit doc **B1** (`motion`'s random
fields), `scripts/nightly_reelcount.py` (the distinct GPU e2e).
25
26     ## 2. Decisions *(D1?D11 **LOCKED** 2026-06-25)*
27
28     | # | Decision (recommendation) | Rationale | Status |
29     |---|-----|-----|-----|
30     | **D1** | **Snapshot the DECISION output, never the encoded reel.** Freeze the rally
segment list + the stitch/concat plan; never MD5 the reel `.mp4`. | Encoded mp4 bytes are non-
deterministic (ffmpeg version string, x264 threading, NVENC, mux timestamps) ? a reel-MD5 false-
alarms on every ffmpeg bump/machine. The repo's own `nightly_reelcount.py` already asserts
```

```

count+metrics, **not** bytes [verified]. | **LOCKED** (2026-06-25) |
31 | **D2** | **Input = a cached trajectory (detector output), not raw video.** Run from a
pinned trajectory CSV (the `fx_r0X` fixtures). | The WASB detector (video?trajectory) is the
GPU/WSL + CV-float-non-deterministic upstream; caching its output makes the proxy deterministic
+ CI-runnable [verified ? the fixtures are post-detector]. | **LOCKED** (2026-06-25) |
32 | **D3** | **Coverage = segmentation decision + stitch/concat plan; STOP before the
encode.** Extend the existing segmentation snapshot down to the concat manifest (clips, cut
points, order, padding). | Segmentation is already snapshotted [verified]; the stitch plan is
the deterministic missing layer. The encode is non-deterministic (D1). | **LOCKED** (2026-06-25)
|
33 | **D4** | **Compare = canonical-JSON snapshot with a readable diff** (an MD5 of the
canonical JSON may be a fast-path summary, but the field-level diff is the artifact). | A
refactor that trips it needs a debuggable diff, not "hash changed" ? mirrors
`test_replay_matches_committed_expected`'s mismatch report [verified]. | **LOCKED** (2026-06-25)
|
34 | **D5** | **Snapshot only deterministic segmenters/fields; exclude the random
fabricated fields.** Use a deterministic segmenter (CONTROL/fusion on trajectory) and a
`{start,end,shuttle}`-style schema that excludes `motion`'s random
`server/receiver/shot_count/avg_speed`. | `motion` fabricates random fields (audit **B1**)
[verified] ? non-snapshot-able as-is. **B1's fix (drop the fabrication) is the enabler that
would make `motion` snapshot-able. | **LOCKED** (2026-06-25) |
35 | **D6** | **The reel mp4 gets a FUNCTIONAL smoke, not a byte hash** (exists, ~duration
within tol, N segments, sane streams). | Corollary of D1 ? verify the encode is *sane*, not
*identical*. | **LOCKED** (2026-06-25) |
36 | **D7** | **Runs in per-PR CI, offline, CONTROL-only ? a refactor gate, not a quality
floor.** | The point is catching a behavior change in a refactor *before merge*; it must be
fast, GPU-free, on the pinned CONTROL config. | **LOCKED** (2026-06-25) |
37 | **D8** | **Intentional changes re-freeze WITH a documented rationale (Launch
Report).** A `--update-snapshot` flag exists, but a mismatch on an *intended* change must be re-
frozen with the Launch-Report rationale recorded. | Without the discipline a snapshot becomes a
rubber stamp; ties to the launch-report convention. | **LOCKED** (2026-06-25) |
38 | **D9** | **Scope boundary: this is the quality-NEUTRAL proxy ? distinct from the M4
quality-FLOOR (per-slug F1), Tier-2 real-video e2e (GPU), and `nightly_reelcount.py`.** Keep
them separate + complementary. | Conflating "behavior unchanged" with "quality is good enough"
muddies both. | **LOCKED** (2026-06-25) |
39 | **D10** | **Deterministic-segmenters-only, ENFORCED by a determinism guard.** Cover
only deterministic segmenters; to *ensure* no non-deterministic one slips in (? a flaky
snapshot), the harness runs each fixture **twice and asserts byte-identical decision output**
before freezing ? any segmenter whose two runs differ (hidden `random`, set-ordering, float/time
non-determinism) is **rejected from snapshot scope** and flagged. Mechanical, not a hand-
maintained allowlist. | Owner, 2026-06-25. `motion` is non-deterministic until **B1 / #385**
lands; then it auto-becomes eligible (no allowlist edit). | **LOCKED** (2026-06-25) |
40 | **D11** | **Auto-sign-off integration is STAGED: ADVISORY first ? GATING later.** Ship
the snapshot as an advisory/informational check; graduate it to a **gating** check (gates a non-
AST-identical refactor's auto-LGTM) only after it's proven stable on real refactors. | Owner,
2026-06-25. Resolves S8 Q1 ? the strategic payoff, de-risked by starting advisory. | **LOCKED**
(2026-06-25) |
41
42 ## 4. Design / architecture
43 **Flow (all offline, deterministic):**
44 `pinned trajectory CSV ? [segmentation: CONTROL segmenter] ? segment list ?
[deterministic stitch-PLAN builder: clips + cut points + padding + order, NO ffmpeg] ? canonical
JSON {schema_version, segments[], stitch_plan{}} ? MD5 / field-diff vs frozen golden`.
45 Mismatch ? behavior changed (decide if intended; re-freeze via D8).
46
47 **Reuse map** [verified unless noted]:
48 - `tests/test_golden_fixtures.py::test_replay_matches_committed_expected` +
`backend/eval/golden_fixtures.py` ? the existing characterization snapshot at the
**segmentation** layer; extend its `expected` schema + replay to also carry the stitch plan.
** (reuse + extend) **
49 - `eval_baselines/fixtures/` (`fx_r0X` + `manifest.json`) ? the pinned post-detector
trajectories. ** (reuse) **
50 - `backend/pipeline/stitcher/compiler.py` ? builds the ffmpeg concat today; ** [design] **
factor out a pure `build_stitch_plan(segments, cfg) -> dict` (the *decision*) from the *encode*,
so the plan is hashable without running ffmpeg. ** (new seam) **

```

```

51     - `eval_baselines/` snapshot store + the cross-OS enforcement already used by the golden
suite. **(reuse)**
52
53     ## 6. Honest limits ? what this does NOT do
54     - Does **not** prove *quality is good* ? only that *behavior is unchanged* vs the frozen
golden (that's M4 / F1-floor's job).
55     - Does **not** cover the **encode** (pixels/bytes) ? only the decision + stitch plan;
the reel gets a functional smoke only (D6).
56     - Does **not** run the **real detector** (GPU/WSL) or real videos ? it runs from cached
trajectories (Tier-2 / `nightly_reelcount.py` cover that, GPU-gated).
57     - A **green snapshot does not bless a behavior change** ? it only certifies *no* change;
an intended change must be re-frozen with a rationale (D8).
58     - Only as good as the **fixtures' coverage** ? pipeline paths not exercised by `fx_r0X`
aren't protected.
59
60     ## 7. Deliverables & progress tracker ? **source of truth**
61     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One small PR per row,
branched from `origin/master`.
62
63     | ID | Deliverable | Depends on | Gated? | Status | PR |
64     |----|-----|-----|-----|-----|----|
65     | P0 | This doc ? decisions in S2 locked | ? | Owner | ? | (this PR) |
66     | P1 | Canonical decision-snapshot schema + deterministic serializer (segments + stitch
plan; excludes non-det fields) | P0 | No | ? | ? |
67     | P2 | Factor `build_stitch_plan()` out of `compiler.py` (pure, ffmpeg-free) + unit test
| P0 | No | ? | ? |
68     | P3 | Extend `golden_fixtures` / `test_golden_fixtures` to capture + compare the stitch
plan **+ the D10 determinism guard (run each fixture twice ? assert byte-identical; reject non-
deterministic segmenters)** | P1, P2 | No | ? | ? |
69     | P4 | Freeze CONTROL snapshots for the existing `fx_r0X` fixtures | P3 | No | ? | ? |
70     | P5 | `--update-snapshot` flow + Launch-Report hook (D8) | P3 | No | ? | ? |
71     | P6 | Wire the snapshot as a per-PR CI characterization gate (CONTROL-only) | P4 | No |
? | ? |
72     | P7 | **Surface the snapshot on refactor PRs ? ADVISORY** (informational; does not
gate) ? per D11 | P6 | No | ? | ? |
73     | P8 | ** (gated) Graduate to GATING auto-sign-off: a snapshot-match marks a non-AST-
identical refactor quality-neutral ? LGTM-eligible (only after P7 proven stable) ? per D11 | P7
| **Owner** | ? | ? |
74
75     ## 8. Open questions ? owner
76     1. ? **RESOLVED (D11):** advisory first ? graduate to gating. *(The strategic payoff,
de-risked by staging.)*
77     2. ? **RESOLVED (D10):** deterministic-segmenters-only, **enforced by the double-run
determinism guard**; **B1 is an enabler, not a blocker** ? and it's being fixed in **#385**,
which makes `motion` auto-eligible.
78     3. ** (still open) Fixture scope:** reuse the existing `fx_r0X` (which stop at
segmentation) by adding a stitch-plan `expected`, or add a dedicated fixture set?
79     4. ? **DONE: D1?D11 LOCKED 2026-06-25.**
80
81     ## 9. Definition of done
82     - [x] D1?D11 locked by the owner (2026-06-25).
83     - [ ] A behavior-preserving refactor passes; a behavior-changing edit fails with a
**readable diff**.
84     - [ ] Runs **offline + deterministically** in per-PR CI, cross-OS (matching the golden
suite).
85     - [ ] Covers **segmentation + stitch plan**; the reel gets a functional smoke only.
86     - [ ] `--update-snapshot` requires a documented Launch-Report rationale.
87     - [ ] `docs/README.md` index + `docs/NEXT_STEPS.md` updated; **$7 all ?** ? flip `Status
? DONE`, archive.
88
89     ## 10. References
90     **Internal code [verified]:** `tests/test_golden_fixtures.py`
(`test_replay_matches_committed_expected`), `backend/eval/golden_fixtures.py`,
`eval_baselines/fixtures/`, `backend/pipeline/stitcher/compiler.py`,
`backend/pipeline/segmenters/motion.py` (B1 random). **Internal docs:**

```

```
`GOLDEN_REGRESSION_FIXTURES.md`, `CODE_CLEANUP_AUDIT_2026-06-24.md` (B1), `docs/NEXT_STEPS.md`  
(M4 quality-floor), `PROJECT_DOC_TEMPLATE.md`; `scripts/nightly_reelcount.py` (distinct GPU  
e2e). **Related:** the AST-identity sign-off used on #379; the Tier-1 nightly (#381).  
91  
92     ---  
93  
94     ### Changelog  
95     - `2026-06-25` ? Initial draft from the "MD5-proxy-for-e2e validation" discussion;  
decisions **D1?D9 proposed for locking**.  
96     - `2026-06-25` ? **Owner LOCKED the decisions.** Added **D11** (auto-sign-off staged:  
advisory ? gating) and **D10** (deterministic-segmenters-only, enforced by a **double-run  
determinism guard** that rejects any segmenter whose two runs differ ? no hand-maintained  
allowlist); D1?D9 locked. **B1 is an enabler, not a blocker** (being fixed in #385 ? `motion`  
becomes snapshot-able). S7 reframed: P7 = advisory, P8 = gating; P3 carries the determinism  
guard.  
97
```

3. user (2026-07-02T02:59:48.625Z)

Found 1 file
badminton-highlight-indexer\docs\DECISION_SNAPSHOT_REGRESSION.md

4. assistant (2026-07-02T02:59:57.633Z)

The doc exists and matches. Now checking git history and the test file claim.

5. user (2026-07-02T03:00:01.145Z)

```
1283e18 docs(project): Decision-Snapshot Regression plan ? a deterministic quality-neutral proof  
(#383)  
---  
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)  
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)  
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)  
aa915a7 deploy/gcp: centralize project + bucket names in one sourced buckets.env (#410)  
52d97b5 P4: POST /api/assets ? store a video into the user's chosen medium (#409)  
---  
1283e18 2026-06-25 docs(project): Decision-Snapshot Regression plan ? a deterministic quality-  
neutral proof (#383)  
  
docs/DECISION_SNAPSHOT_REGRESSION.md | 96 +++++  
docs/README.md                       | 1 +  
2 files changed, 97 insertions(+)  
---  
master
```

6. user (2026-07-02T03:00:01.543Z)

Found 11 files
badminton-highlight-indexer\tests\test_api_endpoints.py
badminton-highlight-indexer\tests\test_watermark_i18n.py
badminton-highlight-indexer\tests\test_watermark.py
badminton-highlight-indexer\tests\test_results.py
badminton-highlight-indexer\tests\test_experiment.py
badminton-highlight-indexer\tests\test_gemini_refine.py
badminton-highlight-indexer\tests\test_generate_highlights.py
badminton-highlight-indexer\tests\test_compute_backend.py
badminton-highlight-indexer\tests\test_compiler.py
badminton-highlight-indexer\tests\test_api_cloud_dispatch.py
badminton-highlight-indexer\tests\test_import_smoke.py

7. assistant (2026-07-02T03:00:11.954Z)

The doc has exactly one commit (1283e18, #383, 2026-06-25) ? never touched since. Let me check

whether the P1/P2 work shipped under a different name.

8. user (2026-07-02T03:00:13.961Z)

Found 2 files

badminton-highlight-indexer\docs\README.md

badminton-highlight-indexer\docs\DECISION_SNAPSHOT_REGRESSION.md

9. user (2026-07-02T03:00:15.181Z)

24:class VideoCompiler:

10. user (2026-07-02T03:00:34.299Z)

```
{"closedAt":null,"mergedAt":null,"state":"OPEN","title":"refactor(D1/P10): extract  
orchestration.py from main.py (god-module split, step 1)"}  
---
```

```
{"mergedAt":"2026-06-25T17:47:44Z","state":"MERGED","title":"docs(project): Decision-Snapshot  
Regression plan ? a deterministic quality-neutral proof"}
```

11. user (2026-07-02T03:01:04.504Z)

Structured output provided successfully